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IT214-006

Happy learning

This diagram shows a school structure. It is designed to show how the school operates and has all the information from the students, school, teachers and parents.



Conceptual Design:

For this diagram the main entities we are creating are the STUDENT, TEACHER, CLASS, PARENT. For our first relationship we have the TEACHER_AT to CLASS_AT Which is many to many (M:M) because one teacher can be in many classes and a class can have many teachers and this relationship is held by a bridge called ASSIGNMENT_AT, which holds FK: CLASS_AT_CODE_AT and TEACHER_AT_ID_AT. Next one is the PARENT_AT to STUDENT_AT which is a many to many (M:M) relationship because students can have more then one parent and parents can have more than one kid and this relationship is held by a bridge called RELATIONSHIP, which is holding FK: STUDENT_AT_ID_AT and PARENT_AT_ID_AT. The last one is the STUDENT_AT to CLASS_AT which is many to one (M:1) because a class can have many kids and kids can only be in one class time.

Entities Analysis:

- STUDNET_AT

- This entity shows the students information table.
- For this entity since there is no PK given we will use a surrogate key ID_AT and the foreign key is CLASS_AT_CODE_AT which comes from CLASS_AT ID CODE_AT.

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
NICK_NAME_AT	VARCHAR(10)	Simple	Single	Required	Duplicated
DATE_OF_BIRTH_AT	DATE	Simple	Single	Required	Duplicated
ADDRESS_AT	VARCHAR(55)	Composite	Single	Required	Duplicated
CITY_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
ZIPCODE_AT	DECIMAL(0,5)	Simple	Single	Required	Duplicated
STREET_AT	VARCHAR(35)	Simple	Single	Required	Duplicated
CLASS_AT_CODE_AT	VARCHAR(2)	Simple	Single	Required	Duplicated

- PARENT_AT

- This entity shows the parent/guardian information table.
- For this entity since there is no PK given we will use a surrogate key ID_AT and there is

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
HOME_PHONE_AT	CHAR(13)	Simple	Single	Optional	Duplicated
MOBILE_PHONE_AT	CHAR(13)	Simple	Single	Required	Duplicated
WORK_PHONE_AT	CHAR(13)	Simple	Single	Optional	Duplicated
EMAIL_AT	VARCHAR(30)	Simple	Single	Optional	Duplicated
RELATION_AT	VARCHAR(20)	Simple	Single	Required	Duplicated

no foreign key

- **CLASS_AT**

- This entity shows the class information table.

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
CLASS_CODE_AT	VARCHAR(2)	Composite	Single	Required	Unique
AGE_LEVEL_AT	VARCHAR(7)	Simple	Single	Required	Duplicated
COLOR_AT	VARCHAR(5)	Simple	Single	Required	Duplicated
ROOM_NUMBER_AT	INT	Simple	Single	Required	Duplicated
PHONE_NUMBER_AT	CHAR(13)	Simple	Single	Required	Duplicated
DIRECTION_AT	VARCHAR(160)	Simple	Single	Optional	Duplicated
CAPACITY_AT	INT	Simple	Single	Required	Duplicated

- For this entity the primary key is the CODE_AT and there is no foreign key

- **TEACHER_AT**

- This entity shows the teacher information table.
 - For this entity since there is no PK given we will use a surrogate key ID_AT and there is no foreign key

Attribute Name	Domain & Size	Simple/ Composite	Single/ Multi-value	Required/ Optional	Unique/ Duplicated
ID_AT	INT	Simple	Single	Required	Unique
FIRST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
LAST_NAME_AT	VARCHAR(15)	Simple	Single	Required	Duplicated
MIDDLE_NAME_AT	VARCHAR(15)	Simple	Single	Optional	Duplicated
HOME_PHONE_AT	CHAR(13)	Simple	Single	Required	Duplicated
WORK_EMAIL_AT	VARCHAR(30)	Simple	Single	Required	Duplicated
PERSONAL_EMAIL_AT	VARCHAR(30)	Simple	Single	Optional	Duplicated
COLLEGE_AT	VARCHAR(50)	Simple	Single	Optional	Duplicated
HIGHEST_DEGREE_AT	VARCHAR(5)	Simple	Single	Required	Duplicated
DEGREE_AREA_AT	VARCHAR(25)	Simple	Single	Required	Duplicated

- **RELATIONSHIP_AT**

- This entity shows the relationship between Student & Parent. Since the relationship between Student and Parent is M:M we create a bridge called **RELATIONSHIP** to connect them.

Attribute Name	Domain & Size
STUDENT_AT_ID_AT	INT
PARENT_AT_ID_AT	INT

- **ASSIGNMENT_AT**

- This entity shows the relationship between Class & Teacher. Since the relationship between Class and Teacher is M:M we create a bridge called **ASSIGNMENT** to connect them.

Attribute Name	Domain & Size
CLASS_AT_CODE_AT	INT
TEACHER_AT_ID_AT	INT

Relationship Analysis:

→ TEACHER_AT to CLASS_AT

- ◆ We use a bridge : **ASSIGNMENT_AT**
 - TEACHER has class ASSIGNMENT (1:M)
 - CLASS has teacher ASSIGNMENT (1:M)
- ◆ M:M because one teacher can be in many classes and one class can have many teachers.
- ◆ The straight is weak because both entities exist independently, not relying on each other.

- ◆ TEACHER(total) because teachers get assigned to a class and CLASS (total) because all classes have teachers.
- ◆ No special cardinality
- ◆ FK: The ASSIGNMENT table holding the CLASS_AT_CODE_AT reference to the class entity and TEACHER_AT_ID_AT reference to the teacher entity

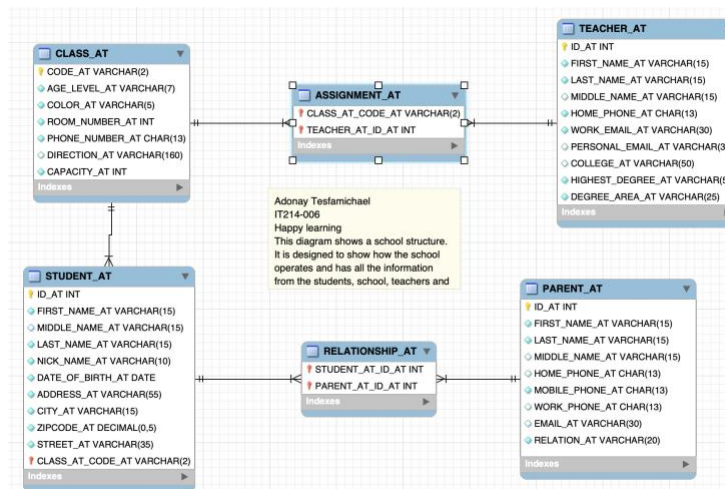
→ PARENT_AT to STUDENT_AT

- ◆ We use a bridge: **RELATIONSHIP_AT**
 - PARENT has student RELATIONSHIP (1:M)
 - STUDENT has parent RELATIONSHIP (1:M)
- ◆ M:M because students can have more than one parent and parents can have more than one kid
- ◆ The strength is strong because each student is connected to a parent or a caretaker
- ◆ They are both partial because they both need each other
- ◆ No special cardinality
- ◆ FK: The RELATIONSHIP table holding the STUDENT_AT_ID_AT reference to the student entity and PARENT_AT_ID_AT reference to the parent entity

→ STUDENT_AT to CLASS_AT

- ◆ STUDENT enrolls in CLASS
- ◆ One CLASS contains many STUDENTS, and one STUDENT is enrolled in one CLASS at a time. No bridge
- ◆ The straight is weak because both entities exist independently, not relying on each other.
- ◆ CLASS (partial) because all the classes might not have kids sometimes, and STUDENTS (total) because a student needs to be always enrolled in a class.
- ◆ No special cardinality
- ◆ FK: CLASS_AT_CODE_AT reference to the CLASS entity

Entity Relationship Diagram:



TEACHER_AT

TEACHER_AT - Table					
Name: TEACHER_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
HOME_PHONE_AT	CHAR(13)	☐	☒	☐	
WORK_EMAIL_AT	VARCHAR(30)	☐	☒	☐	
PERSONAL_EMAIL_AT	VARCHAR(30)	☐	☐	☐	
COLLEGE_AT	VARCHAR(50)	☐	☐	☐	
HIGHEST_DEGREE_AT	VARCHAR(5)	☐	☒	☐	
DEGREE_AREA_AT	VARCHAR(25)	☐	☒	☐	

STUDENT_AT

STUDENT_AT - Table					
Name: STUDENT_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
NICK_NAME_AT	VARCHAR(10)	☐	☒	☐	
DATE_OF_BIRTH_AT	DATE	☐	☒	☐	
ADDRESS_AT	VARCHAR(55)	☐	☒	☐	
CITY_AT	VARCHAR(15)	☐	☒	☐	
ZIPCODE_AT	DECIMAL(0,5)	☐	☒	☐	
STREET_AT	VARCHAR(35)	☐	☒	☐	
CLASS_AT_CODE_AT	VARCHAR(2)	☒	☒	☐	

PARENT_AT

PARENT_AT - Table					
Name: PARENT_AT					
Column	Datatype	PK	NN	UQ	
ID_AT	INT	☒	☒	☒	
FIRST_NAME_AT	VARCHAR(15)	☐	☒	☐	
LAST_NAME_AT	VARCHAR(15)	☐	☒	☐	
MIDDLE_NAME_AT	VARCHAR(15)	☐	☐	☐	
HOME_PHONE_AT	CHAR(13)	☐	☐	☐	
MOBILE_PHONE_AT	CHAR(13)	☐	☒	☐	
WORK_PHONE_AT	CHAR(13)	☐	☐	☐	
EMAIL_AT	VARCHAR(30)	☐	☐	☐	
RELATION_AT	VARCHAR(20)	☐	☒	☐	

CLASS_AT

Name: CLASS_AT					
Column	Datatype		PK	NN	UQ
CODE_AT	VARCHAR(2)	↕	☒	☒	☒
AGE_LEVEL_AT	VARCHAR(7)	↕	☐	☒	☐
COLOR_AT	VARCHAR(5)	↕	☐	☒	☐
ROOM_NUMBER_AT	INT	↕	☐	☒	☐
PHONE_NUMBER_AT	CHAR(13)	↕	☐	☒	☐
DIRECTION_AT	VARCHAR(160)	↕	☐	☐	☐
CAPACITY_AT	INT	↕	☐	☒	☐

BRIDGE: RELATIONSHIP_AT

- (STUDENT & PARENT)

Name: RELATIONSHIP_AT					
Column	Datatype		PK	NN	UQ
STUDENT_AT_ID_AT	INT	↕	☒	☒	☐
PARENT_AT_ID_AT	INT	↕	☒	☒	☐

BRIDGE: ASSIGNMENT_AT

- (CLASS & TEACHER)

Name: ASSIGNMENT_AT					
Column	Datatype		PK	NN	
CLASS_AT_CODE_AT	VARCHAR(2)	↕	☒	☒	
TEACHER_AT_ID_AT	INT	↕	☒	☒	